

Zoonotic, Travel, Fungal, and Parasitic Infections

Medvale Internal Medicine - Infectious Diseases

Purpose of this chapter

This section consolidates the infections that are usually learned as pattern recognition: tick exposure, animal exposure, travel, immunocompromise, pulmonary nodules or cavities, meningitis in AIDS, eosinophilia, chronic diarrhea, and exposure-linked skin findings. The goal is not to memorize every organism, but to connect exposure, host immune state, syndrome, diagnostic test, and first treatment move.

Source integration note. This chapter uses the uploaded Medvale/Step-Up-style source section as a coverage signal for Lyme disease, Rocky Mountain spotted fever, malaria, rabies, candidiasis, aspergillosis, cryptococcosis, and common fungal/parasitic infection categories. The wording, tables, algorithms, and Graph-RAG maps below are original and were cross-checked against current public clinical guidance.

1. Why this matters clinically

Zoonotic, travel-associated, fungal, and parasitic infections are high-yield because the diagnosis is often missed unless the clinician asks a targeted exposure question. A patient with fever, thrombocytopenia, and hyponatremia after camping needs a different mental model than a patient with fever after central-line placement. A traveler with fever after returning from sub-Saharan Africa has malaria until proven otherwise. A patient with advanced HIV, headache, and little meningeal inflammation may have cryptococcal meningitis. A neutropenic patient with persistent fever and pulmonary nodules may have invasive aspergillosis. Each syndrome is a matching problem: host risk plus exposure plus tempo plus organ involvement.

The dangerous traps are delay and false reassurance. Early Lyme disease can be diagnosed clinically without waiting for serology when erythema migrans is present. Rocky Mountain spotted fever can be fatal if doxycycline is delayed while waiting for rash or confirmatory tests. Severe malaria requires immediate IV therapy. Rabies exposure requires wound care and postexposure prophylaxis before symptoms develop because symptomatic rabies is almost universally fatal. Invasive fungal disease often requires treatment before cultures are definitive.

High-yield organizing rule

When an infection does not fit common community-acquired bacterial patterns, immediately ask: Where has the patient been? What animals, ticks, mosquitoes, freshwater, soil, caves, birds/bats, unpasteurized dairy, or occupational exposures occurred? Is the host immunocompromised by HIV, transplant, neutropenia, steroids, diabetes, cirrhosis, or asplenia?

2. Exposure-first diagnostic framework

Exposure / host state	Syndrome clue	Priority diagnoses	First diagnostic move	First management move
Tick exposure in Northeast/Upper Midwest/West Coast	Expanding rash, facial palsy, meningitis, arthritis, AV block	Lyme disease	Clinical diagnosis if classic erythema migrans; two-tier serology for later disease	Doxycycline or amoxicillin/cefuroxime depending on age, pregnancy, and syndrome
Tick exposure with fever, severe headache, thrombocytopenia, hyponatremia, rash starting wrists/ankles	Systemic vasculitic illness	RMSF and other rickettsioses	Clinical diagnosis; serology later	Start doxycycline immediately
Travel to malaria-endemic region	Cyclical or nonspecific fever, anemia, thrombocytopenia, splenomegaly	Malaria, especially <i>P. falciparum</i>	Thick/thin smear or rapid diagnostic test; repeat if negative and suspicion high	Treat immediately; severe disease = IV artesunate
Animal bite from bat, raccoon, skunk, fox, dog, cat	Exposure often asymptomatic initially	Rabies risk	Assess animal availability and exposure category	Wound washing plus rabies immune globulin/vaccine when indicated
Advanced HIV/transplant/steroids	Headache with high opening pressure, pulmonary nodules, fungemia	Cryptococcus, Aspergillus, Candida, endemic fungi	Antigen testing, cultures, imaging, LP when CNS suspected	Targeted antifungal therapy and immune-state management
Cave/bird/bat droppings or Ohio/Mississippi River valleys	Pneumonia, mediastinal LAD, disseminated disease in immunocompromised host	Histoplasmosis	Urine/serum antigen, serology, culture if severe	Itraconazole for mild/moderate; amphotericin for severe disease
Freshwater, contaminated water, flooding, rodents/farm animals	Fever, conjunctival suffusion, jaundice, renal injury	Leptospirosis	Serology/PCR depending on timing	Doxycycline or penicillin/ceftriaxone in severe disease

3. Tick-borne infections

3.1 Lyme disease

Lyme disease is caused by the spirochete *Borrelia burgdorferi*, transmitted by Ixodes ticks. In the United States, the classic regions are the Northeast, upper Midwest, and parts of the West Coast. The vector ecology matters

because immature ticks feed on small mammals and mature ticks feed on deer; humans are incidental hosts. The incubation period is commonly days to weeks, and the clinical course is divided into early localized, early disseminated, and late disease.

Stage	Mechanism	Clinical pattern	Testing logic	Treatment logic
Early localized	Local multiplication at bite site	Erythema migrans: expanding annular lesion, often painless and not always bullseye; fever, malaise, myalgias may occur	Classic erythema migrans in endemic exposure is clinical; early serology may be negative	Oral doxycycline, amoxicillin, or cefuroxime; doxycycline often preferred when coinfection possible
Early disseminated	Hematogenous spread	Multiple lesions, facial nerve palsy, meningitis/radiculopathy, migratory arthralgia, carditis with AV block	Two-tier serology is more sensitive than in early localized disease; ECG if cardiac symptoms	Oral or IV therapy depending neurologic/cardiac severity
Late disease	Persistent immune/inflammatory manifestations	Mono- or oligoarticular arthritis, especially knee; chronic neurologic symptoms less common	Serology usually positive; synovial PCR sometimes helpful	Longer antibiotic courses; evaluate for post-treatment inflammatory arthritis if persistent

Lyme diagnostic pitfall

A negative Lyme serology does not exclude very early erythema migrans. Conversely, IgG can remain positive for years, so a positive test does not prove that current nonspecific symptoms are active infection.

3.2 Rocky Mountain spotted fever and rickettsial illness

Rocky Mountain spotted fever is caused by *Rickettsia rickettsii*, an intracellular organism that infects vascular endothelium. Endothelial injury produces capillary leak, edema, microhemorrhage, and end-organ dysfunction. The classic teaching is fever, headache, and rash beginning on wrists/ankles and spreading centrally, including palms and soles, but rash can be absent early. Waiting for rash or serology is dangerous.

Feature	RMSF teaching anchor
Incubation and tempo	Symptoms often begin about 1 week after tick bite. Early symptoms are nonspecific: fever, severe headache, malaise, myalgias, nausea/vomiting, photophobia.
Rash	Often appears several days after fever. Starts peripherally and spreads centrally; becomes petechial in more severe disease.
Labs	Thrombocytopenia, hyponatremia, elevated transaminases may support the diagnosis.
Management	Doxycycline is first-line for adults and children; start empirically when suspected. Do not wait for confirmatory testing.
Complications	ARDS, renal injury, shock, neurologic injury, limb ischemia/amputation, death.

Rickettsial rule

Doxycycline is the drug to remember for RMSF, ehrlichiosis, anaplasmosis, and many other tick-borne rickettsial syndromes. Early treatment prevents severe illness.

3.3 Other zoonoses and arthropod-borne infections

Disease	Exposure	Illness script	Diagnostic clue	Treatment anchor
Ehrlichiosis / anaplasmosis	Ticks; deer reservoirs	Fever, headache, malaise, cytopenias, transaminitis; rash less prominent than RMSF	Leukopenia/thrombocytopenia; PCR early or serology later	Doxycycline
Tularemia	Ticks, rabbits, animal handling, aerosols	Ulceroglandular disease, fever, painful regional nodes; pneumonic forms possible	Culture requires lab warning; serology often used	Gentamicin or streptomycin for severe disease; doxy/cipro for selected mild cases
Q fever	Farm animals, birth products, aerosols, unpasteurized milk	Atypical pneumonia, hepatitis, fever; chronic endocarditis possible	Serology phase I/II pattern	Doxycycline; chronic disease needs prolonged combination therapy
Cat scratch disease	Cat/kitten scratch; fleas	Regional lymphadenitis, fever; bacillary angiomatosis in AIDS	Clinical plus serology when needed	Often self-limited; azithromycin for significant LAD; broader therapy in immunocompromised hosts

Disease	Exposure	Illness script	Diagnostic clue	Treatment anchor
Leptospirosis	Freshwater/urine from rodents or livestock	Biphasic fever, conjunctival suffusion, jaundice, renal injury, aseptic meningitis	PCR early, serology later	Doxycycline for mild disease; penicillin/ceftriaxone for severe disease
Brucellosis	Unpasteurized dairy, livestock exposure	Undulating fever, sweats, arthralgia, hepatosplenomegaly	Blood cultures/serology	Doxycycline plus rifampin; longer courses for focal disease

4. Malaria and travel-associated fever

Malaria is a protozoal infection transmitted by Anopheles mosquitoes. The clinically important species include Plasmodium falciparum, P. vivax, P. ovale, P. malariae, and P. knowlesi. Fever after travel to an endemic region is malaria until proven otherwise, especially if the patient came from sub-Saharan Africa, the Amazon basin, Oceania, or parts of South/Southeast Asia. The disease can progress quickly, and P. falciparum is the species most associated with severe and fatal disease.

Species / pattern	Key clinical point	Relapse issue	Management implication
P. falciparum	Most dangerous; fever may be continuous or irregular; high parasitemia, cerebral malaria, severe anemia, AKI, ARDS, shock, acidosis possible	No dormant hypnozoite relapse	Hospitalize if falciparum or severe features; severe disease requires IV artesunate
P. vivax / P. ovale	Often periodic fever every 48 hours after synchronization; can still be severe	Dormant liver hypnozoites cause relapse	Treat blood-stage infection plus primaquine/tafenoquine for radical cure after checking G6PD status
P. malariae	Classically 72-hour fever periodicity; lower-grade chronic infection possible	No hypnozoite relapse	Treat blood-stage infection; evaluate for chronic complications
P. knowlesi	Southeast Asia; can replicate every 24 hours and become severe	No hypnozoite relapse	Do not dismiss as benign malaria; monitor closely

Diagnosis. Thick and thin blood smears remain central because they identify parasites, allow species identification, and estimate parasitemia. Rapid diagnostic tests help when microscopy is delayed, but a negative rapid test does not fully exclude malaria when suspicion is high. Repeat smears every 12-24 hours for several sets if initial smears are negative and suspicion remains. CBC often shows anemia and thrombocytopenia; CMP may show bilirubin elevation, transaminitis, renal injury, or acidosis.

Severe malaria. Red flags include altered mental status/coma, severe anemia, renal injury, ARDS, shock, acidosis, DIC, jaundice plus another severe feature, or high parasitemia. Severe disease should be treated urgently with IV artesunate in the United States.

Travel fever triage

Fever after travel is not a single diagnosis. Always ask about itinerary, rural exposure, mosquito/tick bites, freshwater, sexual exposure, animal exposure, food/water, vaccinations, prophylaxis adherence, and timing. Malaria can occur even when chemoprophylaxis was prescribed if adherence was poor or resistance/exposure occurred.

5. Rabies and postexposure logic

Rabies is an encephalitic viral infection transmitted through saliva from infected mammals. Dogs, bats, raccoons, skunks, foxes, and coyotes are classic reservoirs. Once neurologic symptoms appear, rabies is almost invariably fatal, so the clinical decision is made at the exposure stage, not after symptoms develop.

Step	Clinical action	Reasoning
1. Wound care	Immediate vigorous washing with soap and water; irrigate and debride as appropriate	Local viral inoculum reduction is a critical first intervention
2. Animal assessment	Identify species, behavior, availability for observation/testing, local epidemiology, and whether bite was provoked	Healthy dogs/cats/ferrets can sometimes be observed; bats and wild carnivores usually require more caution
3. Passive immunization	Rabies immune globulin infiltrated around wound when indicated, with remaining volume IM at distant site	Provides immediate neutralizing antibody before vaccine response
4. Active immunization	Rabies vaccine series for previously unvaccinated patients; modified schedule for immunocompromised hosts or prior vaccination	Generates durable immune response

Symptoms, when they occur, may begin with paresthesias or pain at the bite site, then progress to anxiety, confusion, hydrophobia, aerophobia, hypersalivation, seizures, coma, or paralytic disease. Prevention and timely postexposure prophylaxis are the clinically important interventions.

6. Fungal infections: host-state framework

Fungal disease is best organized by host state and route of acquisition. Mucocutaneous *Candida* reflects local overgrowth or immune dysfunction; candidemia reflects intravascular access or severe illness. *Aspergillus* is inhaled and becomes dangerous when neutrophil or macrophage defenses are impaired. *Cryptococcus* is associated with pigeon droppings and advanced cellular immune dysfunction. Endemic dimorphic fungi are geographic organisms that can cause pneumonia in immunocompetent hosts but disseminate in immunocompromised hosts.

Organism / syndrome	Usual host-risk pattern	Clinical anchors	Diagnostic anchors	Treatment anchors
<i>Candida</i> mucocutaneous disease	Antibiotics, diabetes, pregnancy, inhaled steroids, HIV	Thrush, esophagitis/odynophagia, vulvovaginitis, intertrigo/satellite lesions	Clinical, KOH budding yeast/pseudohyphae	Topical azoles/nystatin for local disease; fluconazole for esophagitis
Candidemia / invasive candidiasis	Central lines, TPN, abdominal surgery, ICU, neutropenia	Persistent fever, sepsis, endophthalmitis risk, hepatosplenic lesions	Blood cultures; repeat cultures; eye exam in selected patients	Remove infected lines when possible; echinocandin initially then step-down if stable/susceptible
Aspergillosis	Neutropenia, transplant, steroids, advanced lung disease cavities	ABPA in asthma/CF; aspergilloma fungus ball; invasive disease with fever, pleuritic pain, hemoptysis, nodules/halo sign	CT chest, galactomannan/BAL, culture/histology showing septate acute-angle hyphae	Voriconazole for invasive disease; surgery for massive hemoptysis/fungus ball complications
Cryptococcosis	Advanced HIV, transplant, steroids; pigeon droppings	Meningitis with high opening pressure, headache, fever, cranial neuropathies; pulmonary nodules	CSF cryptococcal antigen, India ink less sensitive, culture	Induction antifungal therapy for meningitis; manage intracranial pressure
Histoplasmosis	Ohio/Mississippi River valleys, bat/bird droppings, caves	Pneumonia, mediastinal LAD; disseminated fever, hepatosplenomegaly, pancytopenia in immunocompromised	Urine/serum antigen, serology, culture	Itraconazole for mild/moderate; amphotericin for severe/disseminated disease
Coccidioidomycosis	Southwest US/desert soil	Pneumonia, erythema nodosum, arthralgia; meningitis or dissemination in high-risk hosts	Serology, culture, imaging	Often supportive if mild; azoles for severe, disseminated, or high-risk disease
Blastomycosis	Great Lakes/Ohio/Mississippi regions, soil/wooded waterways	Pneumonia plus verrucous skin lesions or bone disease	Urine antigen cross-reacts; culture/histology broad-based budding yeast	Itraconazole mild/moderate; amphotericin severe/CNS
Sporotrichosis	Rose thorns, sphagnum moss, cats	Nodular lymphangitis tracking up extremity	Culture	Itraconazole

7. Candida in detail

Candida normally colonizes mucosal surfaces. Antibiotic exposure, hyperglycemia, immune dysfunction, moisture, and epithelial disruption allow overgrowth. Mucosal disease is common and usually not invasive; candidemia is a bloodstream infection that should prompt a search for intravascular catheters, abdominal source, endophthalmitis, and persistent metastatic foci.

Syndrome	Clues	Diagnostic approach	Management logic
Oropharyngeal candidiasis	White plaques that scrape off; burning mouth; consider HIV if unexplained	Clinical or KOH	Topical clotrimazole/nystatin if mild; fluconazole if severe/recurrent
Esophageal candidiasis	Odynophagia/dysphagia in HIV, transplant, steroids	Often empiric diagnosis if classic; endoscopy if no response	Systemic fluconazole
Vulvovaginal candidiasis	Pruritus, thick discharge, normal pH, soreness	Wet mount/KOH or NAAT/culture if recurrent	Topical azole or fluconazole; longer regimens for recurrent disease

Syndrome	Clues	Diagnostic approach	Management logic
Candidemia	ICU or central line patient with persistent fever/sepsis	Blood cultures, repeat cultures until clearance	Echinocandin first in many adults; remove infected catheter when feasible

8. Aspergillus in detail

Aspergillus spores are inhaled frequently, but invasive disease is uncommon without impaired innate immunity. Neutropenia and transplant-related immunosuppression are classic risks. The same organism can produce allergic disease, colonization/fungus ball, or invasive angioinvasive disease depending on host defenses and lung architecture.

Form	Host context	Key presentation	Imaging / diagnostic clue	Treatment direction
Allergic bronchopulmonary aspergillosis (ABPA)	Asthma or cystic fibrosis	Recurrent wheeze, mucus plugging, eosinophilia, high IgE	Central bronchiectasis, Aspergillus-specific IgE/IgG	Steroids; antifungal sometimes adjunctive
Aspergilloma	Preexisting lung cavity from TB, sarcoid, bronchiectasis	Often asymptomatic; hemoptysis can be massive	Mobile fungus ball in cavity	Observe if stable; surgery/embolization for severe hemoptysis
Invasive aspergillosis	Neutropenia, transplant, high-dose steroids	Persistent fever, pleuritic chest pain, cough, hemoptysis	CT nodules, halo sign; galactomannan/BAL; tissue invasion	Voriconazole-based therapy; reduce immunosuppression when possible

9. Cryptococcus in detail

Cryptococcus neoformans is a heavily encapsulated yeast associated with pigeon droppings and soil. Disease classically affects patients with impaired T-cell immunity, especially advanced HIV, transplant immunosuppression, or chronic steroids. CNS disease may produce relatively muted CSF inflammation despite high fungal burden, so opening pressure and cryptococcal antigen testing are crucial.

Clinical question	High-yield answer
When to suspect?	Advanced HIV or transplant patient with subacute headache, fever, confusion, cranial neuropathy, or high opening pressure; pulmonary nodules can occur.
Best tests?	Lumbar puncture with opening pressure, CSF cryptococcal antigen, fungal culture. Serum cryptococcal antigen helps in disseminated disease.
What makes it dangerous?	Elevated intracranial pressure, delayed diagnosis, immune reconstitution inflammatory syndrome after ART, fungemia.
Treatment concept?	Induction therapy for CNS disease followed by consolidation and maintenance therapy; control intracranial pressure with repeat LPs when needed.

10. Parasitic infections: syndrome map

Parasitic infections are approached by geography, immune status, stool pattern, eosinophilia, and organ involvement. Protozoa often cause diarrhea, liver abscess, malaria, or CNS disease without eosinophilia. Helminths commonly produce eosinophilia during tissue migration, anemia from blood loss, biliary obstruction, pulmonary symptoms, or chronic abdominal complaints.

Disease	Exposure / path	Clinical anchors	Diagnostic anchors	Treatment anchors
Giardiasis	Contaminated freshwater, daycare, campers	Foul-smelling greasy diarrhea, bloating, flatulence, weight loss	Stool antigen or PCR	Tinidazole, metronidazole, or nitazoxanide
Amebiasis	Contaminated food/water; travel	Dysentery; liver abscess with RUQ pain and fever	Stool antigen/PCR; serology for liver abscess	Metronidazole/tinidazole plus luminal agent
Cryptosporidiosis	Waterborne; AIDS severe chronic diarrhea	Watery diarrhea, biliary disease in AIDS	Stool acid-fast/PCR/antigen	Supportive; nitazoxanide if immunocompetent; ART immune restoration in HIV

Disease	Exposure / path	Clinical anchors	Diagnostic anchors	Treatment anchors
Toxoplasmosis	Cat feces, undercooked meat; latent reactivation in AIDS	Ring-enhancing brain lesions, seizures, focal deficits; congenital infection	Serology; brain imaging; response to therapy	Pyrimethamine-sulfadiazine-leucovorin or alternatives; prophylaxis in advanced HIV when indicated
Strongyloidiasis	Skin penetration in tropics/subtropics; steroids trigger hyperinfection	Abdominal symptoms, larva currens, eosinophilia; hyperinfection with sepsis/meningitis	Serology, stool larvae insensitive; sputum larvae in hyperinfection	Ivermectin; screen before steroids in high-risk patients
Hookworm	Barefoot soil exposure	Iron deficiency anemia, abdominal symptoms, eosinophilia	Stool ova/parasites	Albendazole/mebendazole plus iron when needed
Schistosomiasis	Freshwater exposure in endemic areas	Swimmer itch, Katayama fever, hematuria or portal hypertension depending species	Eggs in urine/stool; serology	Praziquantel
Taenia solium neurocysticercosis	Fecal-oral eggs from human tapeworm carrier	Seizures, cystic brain lesions	Neuroimaging, serology	Albendazole/praziquantel plus steroids and seizure management; avoid simplistic empiric therapy without imaging context

11. Diagnostic algorithms

11.1 Fever after tick exposure

Step	Decision
1	Is there erythema migrans in an endemic exposure? If yes, treat Lyme clinically; do not wait for early serology.
2	Is illness severe with fever/headache, thrombocytopenia, hyponatremia, rash, or transaminitis? Treat empirically with doxycycline for RMSF/ehrlichiosis/anaplasmosis while testing.
3	Are facial palsy, AV block, meningitis, or arthritis present? Evaluate for disseminated Lyme and determine oral versus IV therapy based on organ severity.
4	Consider coinfection if high fever, cytopenias, hemolysis, or persistent symptoms occur in Ixodes regions: Babesia, Anaplasma, Lyme can overlap.

11.2 Immunocompromised host with pulmonary findings

Host/radiology clue	Most likely fungal cause	Next move
Neutropenia/transplant/steroids with nodules/halo sign	Invasive aspergillosis	CT chest, galactomannan/BAL, start mold-active therapy if suspicion high
Advanced HIV with headache and high opening pressure	Cryptococcus	LP with opening pressure and cryptococcal antigen; induction therapy if positive
Central line/TPN/abdominal surgery with candidemia	Candida	Repeat blood cultures, evaluate catheter/source, echinocandin, source control
Ohio/Mississippi River exposure with pneumonia and mediastinal LAD	Histoplasma	Urine/serum antigen, serology; severity-based antifungal choice
Southwest desert exposure with pneumonia and erythema nodosum	Coccidioides	Serology; treat based on severity and dissemination risk

12. Practice questions

Question 1. A 28-year-old hiker from Connecticut has a 9-cm expanding annular rash on the thigh 8 days after a tick bite. He has malaise but no neurologic or cardiac symptoms. Lyme serology is negative. What is the next step?

Answer. Treat clinically for early localized Lyme disease. Early serology can be negative; classic erythema migrans in an endemic exposure is sufficient for diagnosis.

Question 2. A 9-year-old has fever, severe headache, vomiting, thrombocytopenia, mild transaminitis, and a rash on wrists and ankles after a camping trip. What is the most important management principle?

Answer. Start doxycycline immediately for suspected RMSF/rickettsial illness. Do not wait for confirmatory serology.

Question 3. A traveler returns from Ghana with fever, jaundice, thrombocytopenia, creatinine elevation, and parasitemia on smear. What treatment category is this?

Answer. Severe malaria. Treat urgently with IV artesunate and supportive care.

Question 4. A man wakes to find a bat in his room and cannot exclude contact. The bat is not available for testing. What is the clinical logic?

Answer. Rabies exposure may require postexposure prophylaxis after risk assessment because symptomatic disease is nearly fatal. Wound care plus rabies immune globulin and vaccine are used when indicated for previously unvaccinated persons.

Question 5. A neutropenic leukemia patient has persistent fever despite broad-spectrum antibiotics and CT chest shows nodules with surrounding ground-glass halo. What infection is most concerning?

Answer. Invasive aspergillosis. Obtain fungal diagnostics such as BAL/galactomannan when feasible and start mold-active therapy promptly when suspicion is high.

Question 6. An advanced HIV patient has subacute headache, confusion, and elevated opening pressure. CSF shows few inflammatory cells. What test is most useful?

Answer. CSF cryptococcal antigen and fungal culture. Cryptococcus can have muted CSF inflammation despite high organism burden.

Question 7. A traveler with chronic foul-smelling greasy diarrhea after drinking stream water has no fever or blood. Most likely pathogen?

Answer. Giardia. Stool antigen/PCR can confirm; treat with an appropriate antiprotozoal agent.

Question 8. A patient from an endemic region receives high-dose steroids and develops gram-negative sepsis, meningitis, diarrhea, and larvae in sputum. What parasite explains this?

Answer. Strongyloides hyperinfection. Steroids can precipitate disseminated disease; ivermectin is the key therapy.

13. Common pitfalls

- Waiting for Lyme serology in classic erythema migrans; early serology can be negative.
- Delaying doxycycline in suspected RMSF because rash is absent or pregnancy/age is a concern; severe rickettsial disease is time-sensitive.
- Forgetting malaria in any febrile traveler from an endemic area, even when fever is not periodic.
- Using a single negative malaria smear to exclude disease when clinical suspicion remains high.
- Failing to manage intracranial pressure in cryptococcal meningitis; antifungals alone may not address the immediate pressure problem.
- Treating candidemia without source control or repeat cultures.
- Assuming eosinophilia is present in all parasitic disease; protozoal infections generally do not cause the same eosinophilic pattern as tissue-invasive helminths.
- Giving steroids to a patient at risk for Strongyloides without considering screening or empiric treatment in high-risk settings.
- Ignoring geography: Histoplasma, Coccidioides, Blastomyces, and tick-borne diseases are pattern-based diagnoses tied strongly to place.

14. Graph-RAG extraction map

The following extraction map is designed for Medvale graph-RAG ingestion. Each row can become concept nodes, diagnostic-rule edges, treatment edges, and source-context anchors.

Node	Class	Key properties	High-value edges
Lyme disease	Zoonotic/tick-borne bacterial infection	Borrelia burgdorferi; Ixodes tick; erythema migrans; facial palsy; arthritis; AV block; two-tier serology	transmitted_by Ixodes tick; causes erythema migrans; treated_by doxycycline/amoxicillin/cefuroxime; diagnosed_by clinical rash or serology

Node	Class	Key properties	High-value edges
RMSF	Rickettsial infection	<i>Rickettsia rickettsii</i> ; fever/headache/rash; thrombocytopenia; hyponatremia; endothelial injury	requires_early doxycycline; mimics viral syndrome; causes vasculitis and shock
Malaria	Travel/protozoal infection	<i>Plasmodium</i> species; fever; anemia; thrombocytopenia; smear diagnosis; severe criteria	transmitted_by <i>Anopheles</i> ; severe_form treated_by IV artesunate; relapse_species require hypnozoite therapy
Rabies exposure	Viral zoonosis/prevention emergency	Animal bite/bat exposure; encephalitis; hydrophobia; PEP before symptoms	managed_by wound care; prevented_by HRIG/vaccine; symptomatic_disease nearly fatal
Candidemia	Invasive fungal infection	<i>Candida</i> bloodstream infection; central line; ICU; TPN; abdominal surgery	requires source control; treated_by echinocandin; monitored_by repeat cultures
Invasive aspergillosis	Mold infection	Neutropenia/transplant/steroids; nodules; halo sign; septate acute-angle hyphae	risk_in neutropenia; diagnosed_by CT/BAL/galactomannan; treated_by voriconazole
Cryptococcal meningitis	Encapsulated yeast CNS infection	Advanced HIV/transplant; headache; high opening pressure; cryptococcal antigen	requires LP; treated_by induction/consolidation/maintenance; complication intracranial pressure
Giardiasis	Protozoal diarrheal infection	Freshwater; greasy foul diarrhea; bloating	diagnosed_by stool antigen/PCR; treated_by tinidazole/metronidazole/nitazoxanide
Strongyloidiasis	Helminth infection	Autoinfection; steroid hyperinfection; larvae in stool/sputum; sepsis	worsened_by steroids; treated_by ivermectin; associated_with eosinophilia except severe hyperinfection may lose eosinophilia

15. References and clinical cross-checks

Uploaded source section: infection4.pdf, pages covering Lyme disease, RMSF, malaria, rabies, candidiasis, aspergillosis, cryptococcosis, and common fungal/parasitic infection categories.

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CDC. Malaria diagnosis and treatment guidance, including uncomplicated and severe malaria. Accessed 2026.

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